

On the ratio of surjective group homomorphisms to ring homomorphisms between cyclic rings

Priyabrata Mandal^{1,*} and Deep Bhattacharjee²

¹ Department of Mathematics, Bioinformatics and Computer Applications, Maulana Azad National Institute of Technology Bhopal, Bhopal 462003, India; ORCID: [0000-0001-6472-6239](#)

² Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India; ORCID: [0000-0003-0466-750X](#)

* **Correspondence:** Email: priyabrata@manit.ac.in; itsdeep@live.com (D.B.)

Abstract: For $n \mid m$ there are $\varphi(n)$ surjective group homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ and $2^{\omega(n)}$ ring homomorphisms. We show that their ratio $\rho(n)$ is $2^{\varepsilon(n)}$ times the number of squares in $(\mathbb{Z}/n\mathbb{Z})^\times$, where $\varepsilon(n) \in \{-1, 0, 1\}$ is fixed by the power of 2 dividing n , and that for odd n the ring homomorphisms correspond to the subfields of degree at most 2 of $\mathbb{Q}(\zeta_n)$. We give two-term asymptotic formulas, with explicit constants, for the number of $n \leq x$ with $\rho(n) \notin \mathbb{Z}$ and for $\sum_{n \leq x} \rho(n)$, and check them against exact computations up to $2 \cdot 10^{10}$ and 10^{10} , respectively.

Keywords: homomorphisms of cyclic rings; quadratic residues; Selberg–Delange method

Mathematics Subject Classification: 11A15, 11N25, 11N37, 11R18, 13M05

1 Introduction

Let $n \mid m$. Gallian and Van Buskirk [1] showed that there are exactly $2^{\omega(n)}$ ring homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$, where $\omega(n)$ is the number of distinct prime factors of n , and it is elementary that exactly $\varphi(n)$ of the group homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ are surjective. This paper is about the ratio

$$\rho(n) = \frac{\varphi(n)}{2^{\omega(n)}}. \quad (1)$$

It is often an integer ($\rho(35) = 6$, $\rho(120) = 4$) and sometimes not ($\rho(6) = \frac{1}{2}$, $\rho(98) = \frac{21}{2}$). We determine $\rho(n)$ exactly as a group-theoretic quantity, describe the integers where it fails to be integral, and give two-term asymptotic formulas for how often this happens and for the mean value of ρ .

Write $U_n = (\mathbb{Z}/n\mathbb{Z})^\times$ and $U_n^2 = \{u^2 : u \in U_n\}$, so that $|U_n^2|$ is the number of quadratic residues modulo n that are prime to n . For $n = 2^s n'$ with n' odd put

$$\varepsilon(n) = \begin{cases} 0, & s = 0 \text{ or } s = 2, \\ -1, & s = 1, \\ 1, & s \geq 3, \end{cases} \quad (2)$$

and let

$$\mathcal{E} = \{n \equiv 2 \pmod{4} : \text{every prime divisor of } n/2 \text{ is } \equiv 3 \pmod{4}\}, \quad (3)$$

so that $\mathcal{E} = \{2, 6, 14, 18, 22, 38, 42, 46, 54, \dots\}$.

Theorem 1.1. *For every $n \geq 1$,*

$$\rho(n) = 2^{\varepsilon(n)} |U_n^2|.$$

Moreover, $\rho(n)$ is an integer if and only if $n \notin \mathcal{E}$, and $\rho(n) = |U_n^2|$ whenever $n \not\equiv 2 \pmod{4}$ and $8 \nmid n$.

For odd n the identity $|U_n^2| = \varphi(n)/2^{\omega(n)}$ is classical, and Theorem 1.1 is a short consequence of the structure of U_n . We record it because it gives the exact form of $\rho(n)$ for every n , and the factor $2^{\varepsilon(n)}$ is precisely what produces the set $\mathcal{E}$. Our second result explains the count $2^{\omega(n)}$ on the field side. Let $\zeta_n = e^{2\pi i/n}$ and, for an odd prime p , let $p^* = (-1)^{(p-1)/2}p$.

Theorem 1.2. *Let n be odd and $n \mid m$. For a ring homomorphism $f: \mathbb{Z}_m \rightarrow \mathbb{Z}_n$ let $S(f)$ be the set of primes $p \mid n$ with $f(1) \equiv 1 \pmod{p}$, and put $d_f = \prod_{p \in S(f)} p^*$. Then $f \mapsto \mathbb{Q}(\sqrt{d_f})$ is a bijection from the set of ring homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ onto the set of subfields $K \subseteq \mathbb{Q}(\zeta_n)$ with $[K : \mathbb{Q}] \leq 2$. Moreover $\rho(n) = [\mathbb{Q}(\zeta_n) : M_n]$, where M_n is the compositum of these fields.*

The bijection comes from an isomorphism between the Boolean group of idempotents of $\mathbb{Z}/n\mathbb{Z}$ and the group of quadratic characters of U_n (Proposition 4.3). Figure 1 shows it for $n = 105$. For even n the two groups have orders $2^{\omega(n)}$ and $2^{\omega(n)+\varepsilon(n)}$, which is another way of reading Theorem 1.1.

The remaining results are analytic. Let $\mathcal{E}(x) = \#\{n \leq x : n \in \mathcal{E}\}$, let χ_{-4} be the non-principal character modulo 4, and let γ be Euler's constant.

Theorem 1.3. *As $x \rightarrow \infty$,*

$$\mathcal{E}(x) = \frac{Cx}{\sqrt{\log x}} \left(1 + \frac{\kappa}{\log x} + O\left(\frac{1}{(\log x)^2}\right) \right),$$

where

$$C = \frac{1}{\pi\sqrt{2}} \prod_{p \equiv 3(4)} \left(1 - \frac{1}{p^2}\right)^{-1/2} = 0.24325\,99441\,92945\dots,$$

$$\kappa = \frac{1}{2} - \frac{\gamma}{4} + \frac{\log 2}{4} + \frac{1}{2} \sum_{p \equiv 3(4)} \frac{\log p}{p^2 - 1} + \frac{1}{4} \frac{L'(1, \chi_{-4})}{L(1, \chi_{-4})} = 0.70475\,34517\,05947\dots$$

Theorem 1.4. *As $x \rightarrow \infty$,*

$$\sum_{n \leq x} \rho(n) = \frac{C_2 x^2}{\sqrt{\log x}} \left(1 + \frac{\kappa_2}{\log x} + O\left(\frac{1}{(\log x)^2}\right) \right),$$

where

$$C_2 = \frac{1}{2\sqrt{\pi}} \prod_p \left(1 + \frac{1}{2p}\right) \left(1 - \frac{1}{p}\right)^{1/2} = 0.22909\,07914\,88938\dots,$$

$$\kappa_2 = \frac{1}{4} - \frac{\gamma}{4} - \frac{1}{4} \sum_p \frac{\log p}{(2p+1)(p-1)} = 0.02482\,64240\,33851\dots$$

Both main terms fit into known results. The set $\mathcal{E}$ is twice the set of odd integers composed of primes $\equiv 3 \pmod{4}$. Counting problems of this kind go back to Landau's work on sums of two squares, and Moree [2] showed that such counting functions have asymptotic expansions in powers of $1/\log x$; the leading constant $2C$ for our set appears explicitly in the work of Jones and Zvonkin [3]. Finch and Sebah [4] obtained the leading term of $\sum_{n \leq x} |U_n^2|$; since $\rho(n)$ and $|U_n^2|$ differ only by the factor $2^{\varepsilon(n)}$, their constant is $\frac{43}{40}C_2$. What we add is closed forms for the second-order coefficients κ and κ_2 , self-contained proofs of both two-term expansions by the Selberg–Delange method [5, Theorem II.5.2], and a comparison with exact computations of $\mathcal{E}(x)$ up to $x = 2 \cdot 10^{10}$ and of $\sum_{n \leq x} \rho(n)$ up to $x = 10^{10}$.

Section 2 collects the counting lemmas. Theorem 1.1 is proved in Section 3, Theorem 1.2 in Section 4, and Theorems 1.3 and 1.4 in Section 5, which ends with the numerical comparison.

2 Preliminaries

Ring homomorphisms are not required to send 1 to 1; this is the convention of [1].¹ We write $B_n = \{e \in \mathbb{Z}/n\mathbb{Z} : e^2 = e\}$ for the set of idempotents.

Lemma 2.1. *Let $n \mid m$. The map $f \mapsto f(1)$ is a bijection from the ring homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ onto B_n , and $|B_n| = 2^{\omega(n)}$.*

Proof. An additive map satisfies $f(k) = kf(1)$, and multiplicativity gives $f(1) = f(1)^2$. Conversely, if $e \in B_n$ then $k \mapsto ke$ is well defined on $\mathbb{Z}_m$ because $n \mid m$, it is additive, and $(ab)e = (ab)e^2 = (ae)(be)$. Write $n = \prod_{i=1}^k p_i^{r_i}$. By the Chinese remainder theorem $\mathbb{Z}/n\mathbb{Z} \cong \prod_i \mathbb{Z}/p_i^{r_i}\mathbb{Z}$, and each factor is a local ring, whose only idempotents are 0 and 1. Hence $|B_n| = 2^k$. $\square$

The proof shows more. The map $e \mapsto S(e) = \{p_i : e \equiv 1 \pmod{p_i}\}$ is a bijection from B_n onto the subsets of $\{p_1, \dots, p_k\}$; note that $e \equiv 1 \pmod{p_i}$ forces $e \equiv 1 \pmod{p_i^{r_i}}$, since $e \pmod{p_i^{r_i}} \in \{0, 1\}$. With the Boolean sum $e \oplus e' = e + e' - 2ee'$, which reduces to addition modulo 2 in each component, B_n is an elementary abelian 2-group and

$$S(e \oplus e') = S(e) \triangle S(e'). \quad (4)$$

Lemma 2.2. *Let $n \mid m$. Exactly $\varphi(n)$ group homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ are surjective.*

Proof. Group homomorphisms are $k \mapsto ka$ with $a \in \mathbb{Z}_n$ (well defined since $n \mid m$), and $k \mapsto ka$ is onto exactly when $\gcd(a, n) = 1$. $\square$

Lemma 2.3. *For every $n \geq 1$, $[U_n : U_n^2] = 2^{\omega(n) + \varepsilon(n)}$.*

Proof. For a finite abelian group A the squaring map has kernel $A[2]$, so $[A : A^2] = |A[2]|$, and a cyclic group contributes a factor 2 to $|A[2]|$ if its order is even and 1 otherwise. By the Chinese remainder theorem $U_n \cong \prod_{p^r \parallel n} U_{p^r}$. For odd p the group U_{p^r} is cyclic of even order $p^{r-1}(p-1)$; moreover $U_2 = 1$, $U_4 \cong \mathbb{Z}/2$, and $U_{2^s} \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{s-2}$ for $s \geq 3$ [6, Chapter 4]. If $n = 2^s n'$ with n' odd, this gives $[U_n : U_n^2] = 2^{\omega(n') + t}$ with $t = 0, 0, 1, 2$ for $s = 0, s = 1, s = 2, s \geq 3$. Since $\omega(n) = \omega(n') + [s \geq 1]$, the exponent is $\omega(n) + \varepsilon(n)$ in every case. $\square$

3 The exact value of $\rho(n)$

Proof of Theorem 1.1. By Lemma 2.3, $\varphi(n) = |U_n| = |U_n^2| [U_n : U_n^2] = |U_n^2| 2^{\omega(n) + \varepsilon(n)}$. Dividing by $2^{\omega(n)}$ gives $\rho(n) = 2^{\varepsilon(n)} |U_n^2|$.

If $\varepsilon(n) \geq 0$, then $\rho(n)$ is an integer and $n \notin \mathcal{E}$. Let $\varepsilon(n) = -1$, that is $n = 2n'$ with n' odd. Then $U_n \cong U_{n'}$, so $\rho(n) = \frac{1}{2} |U_{n'}^2|$ with

$$|U_{n'}^2| = \prod_{p^r \parallel n'} \frac{p^{r-1}(p-1)}{2}$$

by Lemma 2.3 applied to each cyclic factor. A factor $p^{r-1}(p-1)/2$ is even exactly when $p \equiv 1 \pmod{4}$. So $\rho(n) \in \mathbb{Z}$ if and only if some prime divisor of n' is $\equiv 1 \pmod{4}$, which means $n \notin \mathcal{E}$. For $n = 2$ the product is empty, so $\rho(2) = \frac{1}{2}$, and indeed $2 \in \mathcal{E}$. The last assertion is the case $\varepsilon(n) = 0$. $\square$

Equivalently, $2^{\omega(n)} \mid \varphi(n)$ if and only if $n \notin \mathcal{E}$. Table 1 shows each case. Note that $\rho(n)$ and $|U_n^2|$ really do differ when $n \equiv 2 \pmod{4}$ or $8 \mid n$: for $n = 26$ the ratio is 3 while there are 6 unit squares, and for $n = 120$ the ratio is 4 while there are 2.

¹If one insists on $f(1) = 1$, there is exactly one ring homomorphism $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ when $n \mid m$, and the question studied here disappears.

Table 1. The quantities of Theorem 1.1 for some small n .

n	factorisation	$\varepsilon(n)$	$\varphi(n)$	$2^{\omega(n)}$	$ U_n^2 $	$\rho(n)$	$n \in \mathcal{E}$
2	2	-1	1	2	1	1/2	yes
6	$2 \cdot 3$	-1	2	4	1	1/2	yes
10	$2 \cdot 5$	-1	4	4	2	1	no
12	$2^2 \cdot 3$	0	4	4	1	1	no
24	$2^3 \cdot 3$	1	8	4	1	2	no
26	$2 \cdot 13$	-1	12	4	6	3	no
35	$5 \cdot 7$	0	24	4	6	6	no
45	$3^2 \cdot 5$	0	24	4	6	6	no
98	$2 \cdot 7^2$	-1	42	4	21	21/2	yes
105	$3 \cdot 5 \cdot 7$	0	48	8	6	6	no
120	$2^3 \cdot 3 \cdot 5$	1	32	8	2	4	no

4 The cyclotomic picture

Let $G_n = \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$. The map $u \mapsto \sigma_u$, $\sigma_u(\zeta_n) = \zeta_n^u$, is an isomorphism $U_n \rightarrow G_n$ [7, Chapter 2], and we identify the two groups. Let $X_n = \text{Hom}(U_n, \{\pm 1\})$ be the group of characters of order at most 2. Since U_n/U_n^2 is an elementary abelian 2-group, $|X_n| = [U_n : U_n^2]$, and $\chi \mapsto \ker \chi$ is a bijection from X_n onto the subgroups of index at most 2 in U_n .

Definition 4.1. M_n is the fixed field of U_n^2 in $\mathbb{Q}(\zeta_n)$.

Proposition 4.2. Let $n \geq 1$ and put $w = \omega(n) + \varepsilon(n)$.

- (i) A subfield $K \subseteq \mathbb{Q}(\zeta_n)$ satisfies $K \subseteq M_n$ if and only if $\text{Gal}(K/\mathbb{Q})$ has exponent at most 2. In particular M_n is the compositum of all quadratic subfields of $\mathbb{Q}(\zeta_n)$.
- (ii) $[M_n : \mathbb{Q}] = 2^w$, $[\mathbb{Q}(\zeta_n) : M_n] = |U_n^2|$, and $\mathbb{Q}(\zeta_n)$ has exactly $2^w - 1$ quadratic subfields.
- (iii) $M_n = \mathbb{Q}(\{\sqrt{p^*} : p \mid n, p \text{ odd}\} \cup \{\sqrt{-1} : 4 \mid n\} \cup \{\sqrt{2} : 8 \mid n\})$.

Proof. (i) Every subfield K is Galois over $\mathbb{Q}$, and $\text{Gal}(K/\mathbb{Q}) \cong U_n/H$ where H fixes K . This quotient has exponent at most 2 if and only if $H \supseteq U_n^2$, that is $K \subseteq M_n$. In the elementary abelian 2-group U_n/U_n^2 the trivial subgroup is the intersection of the subgroups of index 2, so U_n^2 is the intersection of the subgroups of index 2 of U_n that contain it. By the Galois correspondence, M_n is the compositum of the quadratic subfields of $\mathbb{Q}(\zeta_n)$.

(ii) The degrees follow from Lemma 2.3 and the Galois correspondence. The quadratic subfields correspond to subgroups of index 2 containing U_n^2 , that is to the $2^w - 1$ non-trivial elements of X_n .

(iii) Let $a_1, \dots, a_t$ be the listed radicands. Each $\sqrt{a_j}$ lies in $\mathbb{Q}(\zeta_n)$: for odd $p \mid n$ the Gauss sum $g_p = \sum_{a=1}^{p-1} \left(\frac{a}{p}\right) \zeta_p^a$ satisfies $g_p^2 = p^*$ [6, Proposition 6.3.2]; $\sqrt{-1} = \zeta_4$; and $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$. Counting the radicands gives $t = \omega(n') + [4 \mid n] + [8 \mid n] = w$, where n' is the odd part of n . The $2^w - 1$ numbers $\prod_{j \in J} a_j$ with $\emptyset \neq J \subseteq \{1, \dots, t\}$ are pairwise distinct modulo $\mathbb{Q}^{\times 2}$. Indeed, a quotient of two of them is, up to a square, again a product over a non-empty set of radicands, and such a product is never a square: an odd prime in it occurs to the first power, and without odd primes it is -1 , 2 or -2 . So they define $2^w - 1$ distinct quadratic subfields of $\mathbb{Q}(\zeta_n)$, which by (ii) are all of them. By (i) their compositum is M_n , and it equals $\mathbb{Q}(\sqrt{a_1}, \dots, \sqrt{a_t})$. $\square$

We call M_n the maximal multiquadratic subfield of $\mathbb{Q}(\zeta_n)$.² For odd p and $u \in U_n$,

$$\sigma_u(g_p) = \sum_{a=1}^{p-1} \left(\frac{a}{p}\right) \zeta_p^{ua} = \left(\frac{u}{p}\right) \sum_{a=1}^{p-1} \left(\frac{ua}{p}\right) \zeta_p^{ua} = \left(\frac{u}{p}\right) g_p. \quad (5)$$

²For odd n with k distinct prime factors, $[M_n : \mathbb{Q}] = 2^k$, so M_n is biquadratic only when $k = 2$.

From now on let n be odd. For $e \in B_n$ define

$$\chi_e(u) = \prod_{p \in S(e)} \left(\frac{u}{p}\right) \quad (u \in U_n), \quad d_e = \prod_{p \in S(e)} p^*, \quad \gamma_e = \prod_{p \in S(e)} g_p,$$

so that $\chi_0 = 1$, $d_0 = 1$, $\gamma_0 = 1$.

Proposition 4.3. *Let n be odd. Then $e \mapsto \chi_e$ is a group isomorphism $(B_n, \oplus) \rightarrow X_n$, and for every $e \in B_n$ the fixed field of $\ker \chi_e$ is $\mathbb{Q}(\sqrt{d_e})$.*

Proof. Each χ_e is a product of characters of order dividing 2, so $\chi_e \in X_n$. By (4), $\chi_{e \oplus e'} = \chi_e \chi_{e'}$, because a prime in $S(e) \cap S(e')$ contributes $(\frac{u}{p})^2 = 1$. If $e \neq 0$, choose $p \in S(e)$, a quadratic non-residue v modulo p , and $u \in U_n$ with $u \equiv v \pmod{p}$ and $u \equiv 1$ modulo every other prime power dividing n ; then $\chi_e(u) = -1$. Hence the homomorphism is injective. Both groups have order $2^{\omega(n)}$, by Lemma 2.1 and by $|X_n| = [U_n : U_n^2]$ with Lemma 2.3, so it is bijective.

By (5), $\sigma_u(\gamma_e) = \chi_e(u)\gamma_e$, and $\gamma_e^2 = d_e$. Hence σ_u fixes $\mathbb{Q}(\gamma_e) = \mathbb{Q}(\sqrt{d_e})$ exactly when $u \in \ker \chi_e$, and the Galois correspondence gives the claim. $\square$

Proof of Theorem 1.2. Compose three bijections: $f \mapsto f(1)$ from ring homomorphisms onto B_n (Lemma 2.1); $e \mapsto \ker \chi_e$ from B_n onto the subgroups of index at most 2 in U_n (Proposition 4.3); and the Galois correspondence from these subgroups onto the subfields of degree at most 2. By Proposition 4.3 the composite sends f to $\mathbb{Q}(\sqrt{d_{f(1)}}) = \mathbb{Q}(\sqrt{d_f})$. Finally, $\varepsilon(n) = 0$ for odd n , so Theorem 1.1 and Proposition 4.2(ii) give $\rho(n) = |U_n^2| = [\mathbb{Q}(\zeta_n) : M_n]$. $\square$

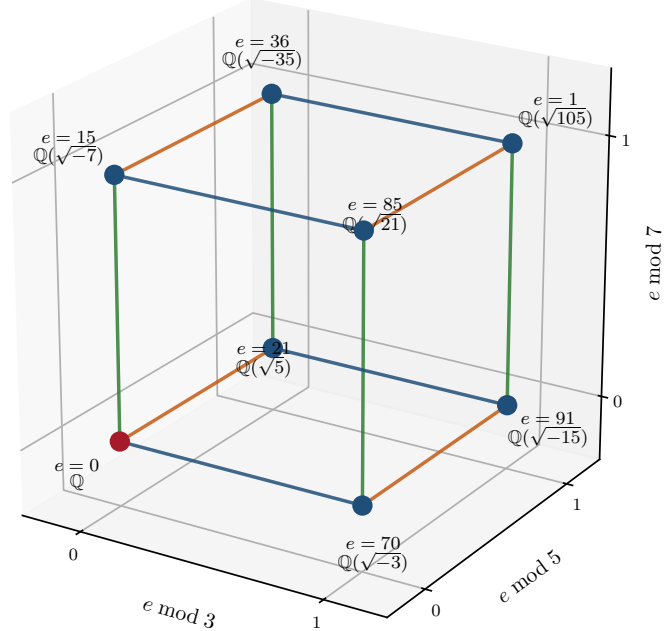


Figure 1. Theorem 1.2 for $n = 105$. The eight idempotents e of $\mathbb{Z}/105\mathbb{Z}$ sit at the vertices ($e \bmod 3$, $e \bmod 5$, $e \bmod 7$) of the cube, and each is labelled with the field $\mathbb{Q}(\sqrt{d_e}) \subseteq \mathbb{Q}(\zeta_{105})$. Moving along an edge changes one coordinate, which is the Boolean sum with 70, 21 or 15 (blue, orange and green edges), and multiplies d_e by -3 , 5 or -7 modulo squares. The compositum of the seven quadratic fields is $M_{105} = \mathbb{Q}(\sqrt{-3}, \sqrt{5}, \sqrt{-7})$, and $[\mathbb{Q}(\zeta_{105}) : M_{105}] = 48/8 = 6 = \rho(105)$.

Remark 4.4. For even n the groups B_n and X_n have orders $2^{\omega(n)}$ and $2^{\omega(n)+\varepsilon(n)}$. When $4 \parallel n$ they are still isomorphic: the 2-component of e is matched with the character $u \mapsto (-1)^{(u-1)/2}$, whose field is $\mathbb{Q}(\sqrt{-1})$. When $n \equiv 2 \pmod{4}$ the 2-component of e has no character to go to, because U_2 is trivial, and when $8 \mid n$ there are two independent characters of U_{2^s} and only one idempotent component. This mismatch is the factor $2^{\varepsilon(n)}$ in Theorem 1.1.

Figure 2 shows the same structure inside a single unit group. With generators 2 of U_5 and 3 of U_7 , the group $U_{35} \cong \mathbb{Z}/4 \times \mathbb{Z}/6$ is drawn on a torus. The two characters $(\frac{\cdot}{5})$ and $(\frac{\cdot}{7})$ split it into four cosets of U_{35}^2 , and the coset of squares has $6 = \rho(35)$ elements.

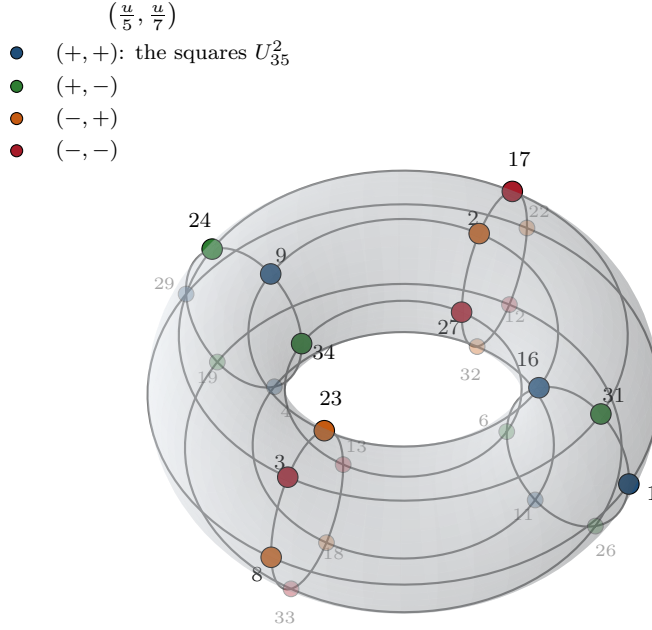


Figure 2. The 24 elements of U_{35} on a torus: u is placed at angle $2\pi k/4$ around the core circle and $2\pi l/6$ around the tube, where $u \equiv 2^k \pmod{5}$ and $u \equiv 3^l \pmod{7}$. The colour records $((\frac{u}{5}), (\frac{u}{7}))$. The six blue points $\{1, 4, 9, 11, 16, 29\}$ form U_{35}^2 , the subgroup fixing $M_{35} = \mathbb{Q}(\sqrt{5}, \sqrt{-7})$. Labels of points on the far side of the torus are drawn faint.

5 Distribution

Write $s = \sigma + i\tau$. We use the Selberg–Delange method in the following form [5, Theorem II.5.2], specialised to non-negative coefficients and two terms.

Lemma 5.1. *Let $a_n \geq 0$ and let $F(s) = \sum_{n \geq 1} a_n n^{-s}$ converge for $\sigma > 1$. Suppose there are $z > 0$, $c_0 > 0$, $0 < \delta \leq 1$ and $M > 0$ such that $G(s) = F(s)\zeta(s)^{-z}$ extends holomorphically to the region $D = \{\sigma \geq 1 - c_0/(1 + \log(1 + |\tau|))\}$ and satisfies $|G(s)| \leq M(1 + |\tau|)^{1-\delta}$ there. Put $Z(s) = ((s-1)\zeta(s))^z G(s)/s$. Then, for $x \geq 3$,*

$$\sum_{n \leq x} a_n = x(\log x)^{z-1} \left(\lambda_0 + \frac{\lambda_1}{\log x} + O\left(\frac{1}{(\log x)^2}\right) \right), \quad \lambda_0 = \frac{Z(1)}{\Gamma(z)}, \quad \lambda_1 = \frac{Z'(1)}{\Gamma(z-1)}.$$

For $z = \frac{1}{2}$ we have $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$. Since $(s-1)\zeta(s) = 1 + \gamma(s-1) + O((s-1)^2)$,

$$\frac{Z'(1)}{Z(1)} = \frac{\gamma}{2} + \frac{G'(1)}{G(1)} - 1, \quad \text{so} \quad \frac{\lambda_1}{\lambda_0} = -\frac{1}{2} \left(\frac{\gamma}{2} + \frac{G'(1)}{G(1)} - 1 \right) \quad \left(z = \frac{1}{2} \right). \quad (6)$$

5.1 The exceptional set

Let $\mathcal{A}$ be the set of odd $a \geq 1$ all of whose prime factors are $\equiv 3 \pmod{4}$. By (3), $\mathcal{E} = \{2a : a \in \mathcal{A}\}$, so $\mathcal{E}(x) = A(x/2)$ with $A(y) = \#\{a \leq y : a \in \mathcal{A}\}$. For $\sigma > 1$ put

$$P(s) = \sum_{a \in \mathcal{A}} a^{-s} = \prod_{p \equiv 3 \pmod{4}} (1 - p^{-s})^{-1}, \quad L(s) = L(s, \chi_{-4}).$$

Comparing Euler factors at $p = 2$, at $p \equiv 1 \pmod{4}$ and at $p \equiv 3 \pmod{4}$ gives, for $\sigma > 1$,

$$P(s)^2 = \frac{(1 - 2^{-s}) \zeta(s) P(2s)}{L(s)}. \quad (7)$$

Indeed, at $p \equiv 3 \pmod{4}$ the right side has factor $(1 - p^{-s})^{-1} (1 + p^{-s}) (1 - p^{-2s})^{-1} = (1 - p^{-s})^{-2}$, and at the other primes all factors cancel. Iterating (7) at $s = 2, 4, 8, \dots$ gives a rapidly convergent formula,

$$P(2) = \prod_{j \geq 1} \left(\frac{(1 - 2^{-2^j}) \zeta(2^j)}{L(2^j)} \right)^{2^{-j}} = 1.16807\,55854\,10514\dots, \quad (8)$$

which is how the value of C was computed.³

Proof of Theorem 1.3. By (7), $G(s) = P(s)\zeta(s)^{-1/2}$ satisfies, for $\sigma > 1$,

$$G(s)^2 = R(s) := \frac{(1 - 2^{-s}) P(2s)}{L(s)}.$$

The product $P(2s)$ converges absolutely and has no zeros for $\sigma > \frac{1}{2}$, and $1 - 2^{-s} \neq 0$ for $\sigma > 0$. Since the modulus is fixed, there is $c > 0$ such that $L(s) \neq 0$ and $1/L(s) \ll \log(|\tau| + 3)$ for $\sigma \geq 1 - c/\log(|\tau| + 3)$ [8, Chapter 11]. Choose $c_0 > 0$ so small that the region D of Lemma 5.1 lies inside this zero-free region and inside $\sigma > \frac{3}{4}$. Then R is holomorphic and zero-free on D , and D is simply connected, so R has a holomorphic square root on D ; we take the one that agrees with G on $\sigma > 1$ (two continuous square roots of the same zero-free function on a connected set agree once they agree at one point). On D we have $|R(s)| \ll \log(|\tau| + 3)$, so $|G(s)| \ll (1 + |\tau|)^{1/2}$ and Lemma 5.1 applies with $z = \delta = \frac{1}{2}$.

Since $L(1) = \pi/4$ and $R(\sigma) > 0$ for real $\sigma \geq 1$,

$$Z(1) = G(1) = \sqrt{R(1)} = \left(\frac{2P(2)}{\pi} \right)^{1/2}, \quad \frac{G'(1)}{G(1)} = \frac{1}{2} \frac{R'(1)}{R(1)} = \frac{1}{2} \left(\log 2 - 2\Sigma_3 - \frac{L'(1)}{L(1)} \right),$$

where $\Sigma_3 = \sum_{p \equiv 3 \pmod{4}} \log p / (p^2 - 1) = -P'(2)/P(2)$. Lemma 5.1 gives

$$A(y) = \frac{y}{\sqrt{\log y}} \left(\lambda_0 + \frac{\lambda_1}{\log y} + O\left(\frac{1}{(\log y)^2} \right) \right), \quad \lambda_0 = \frac{\sqrt{2P(2)}}{\pi},$$

with λ_1/λ_0 given by (6). Put $y = x/2$. Since $(\log y)^{-1/2} = (\log x)^{-1/2} (1 + \frac{\log 2}{2 \log x} + O((\log x)^{-2}))$ and $(\log y)^{-1} = (\log x)^{-1} + O((\log x)^{-2})$,

$$\mathcal{E}(x) = \frac{\lambda_0}{2} \frac{x}{\sqrt{\log x}} \left(1 + \frac{\lambda_1/\lambda_0 + \frac{1}{2} \log 2}{\log x} + O\left(\frac{1}{(\log x)^2} \right) \right).$$

³The first factor involves $L(2)$, which is Catalan's constant. A direct product over the primes $p < 10^7$ agrees with (8) to eight decimal places, as the size of the omitted tail predicts.

Here $\lambda_0/2 = \sqrt{P(2)}/(\pi\sqrt{2}) = C$, and by (6)

$$\frac{\lambda_1}{\lambda_0} + \frac{\log 2}{2} = \frac{1}{2} - \frac{\gamma}{4} - \frac{1}{4} \left(\log 2 - 2\Sigma_3 - \frac{L'(1)}{L(1)} \right) + \frac{\log 2}{2} = \kappa. \quad \square$$

For the numerical value of κ we used $\Sigma_3 = 0.22873\,63531\,94032\dots$. Iterating (7) gives

$$\log P(s) = \sum_{j \geq 0} 2^{-j-1} \log \tilde{R}(2^j s), \quad \tilde{R}(s) = \frac{(1 - 2^{-s}) \zeta(s)}{L(s)},$$

hence the rapidly convergent series $\Sigma_3 = -\frac{1}{2} \sum_{j \geq 1} \tilde{R}'(2^j)/\tilde{R}(2^j)$. We also used

$$\frac{L'(1, \chi_{-4})}{L(1, \chi_{-4})} = \gamma + 2 \log 2 + 3 \log \pi - 4 \log \Gamma\left(\frac{1}{4}\right) = 0.24560\,95847\,77314\dots,$$

which follows from Lerch's formula for $L'(0, \chi_{-4})$ and the functional equation.⁴

5.2 The mean value of ρ

Put $g(n) = \rho(n)/n$. Since $\varphi(p^k)/p^k = 1 - 1/p$, the function g is multiplicative with $g(p^k) = \frac{1}{2}(1 - 1/p)$ for every prime p and $k \geq 1$, including $p = 2$. Its Dirichlet series is, for $\sigma > 1$,

$$F(s) = \sum_{n \geq 1} \frac{g(n)}{n^s} = \prod_p \left(1 + \frac{1 - p^{-1}}{2} \frac{p^{-s}}{1 - p^{-s}} \right) = \prod_p \frac{1 - \frac{1+p^{-1}}{2} p^{-s}}{1 - p^{-s}}. \quad (9)$$

Lemma 5.2. *Let $b > 0$ and $I_b(x) = \int_3^x t(\log t)^{-b} dt$. Then $I_b(x) \ll_b x^2(\log x)^{-b}$ and*

$$I_b(x) = \frac{x^2}{2(\log x)^b} \left(1 + \frac{b}{2 \log x} + O_b\left(\frac{1}{(\log x)^2}\right) \right).$$

Proof. Split at $\sqrt{x}$: the part over $[3, \sqrt{x}]$ is at most $x/2$, since $\log t \geq \log 3 > 1$, and the part over $[\sqrt{x}, x]$ is at most $2^{b-1}x^2(\log x)^{-b}$. This proves the bound. Integration by parts gives $I_b(x) = \frac{1}{2}x^2(\log x)^{-b} + \frac{b}{2}I_{b+1}(x) + O(1)$; applying this once more to I_{b+1} and the bound to I_{b+2} gives the expansion. $\square$

Proof of Theorem 1.4. By (9), $G(s) = F(s)\zeta(s)^{-1/2} = \prod_p E_p(s)$ with

$$E_p(s) = \left(1 - \frac{1 + p^{-1}}{2} u \right) (1 - u)^{-1/2}, \quad u = p^{-s},$$

where $(1 - u)^{-1/2}$ is the principal branch. For $\sigma \geq \frac{3}{4}$ we have $|u| \leq 2^{-3/4}$, and expanding $(1 - u)^{-1/2} = 1 + \frac{u}{2} + O(|u|^2)$ gives $E_p(s) = 1 - \frac{u}{2p} + O(|u|^2) = 1 + O(p^{-7/4} + p^{-3/2})$ uniformly. So the product converges absolutely and uniformly on $\sigma \geq \frac{3}{4}$, and G is holomorphic and bounded there. Lemma 5.1 applies with $z = \frac{1}{2}$, $\delta = 1$ and any c_0 for which $D \subseteq \{\sigma > \frac{3}{4}\}$, and gives

$$S(t) := \sum_{n \leq t} g(n) = \frac{t}{\sqrt{\log t}} \left(\lambda_0 + \frac{\lambda_1}{\log t} \right) + O\left(\frac{t}{(\log t)^{5/2}} \right) \quad (t \geq 3), \quad (10)$$

with $\lambda_0 = G(1)/\sqrt{\pi}$. At $s = 1$, $E_p(1) = (1 + \frac{1}{2p})(1 - \frac{1}{p})^{1/2}$, and a short computation gives

$$\left. \frac{d}{ds} \log E_p(s) \right|_{s=1} = \log p \left(\frac{p+1}{(2p+1)(p-1)} - \frac{1}{2(p-1)} \right) = \frac{\log p}{2(2p+1)(p-1)},$$

⁴We also differentiated $\log L(s, \chi_{-4})$ numerically at $s = 1$; the two values agree to ten decimal places.

so $G'(1)/G(1) = \frac{1}{2} \sum_p \log p / ((2p+1)(p-1))$.

By partial summation, $\sum_{n \leq x} \rho(n) = \sum_{n \leq x} n g(n) = xS(x) - \int_1^x S(t) dt$. On $[1, 3]$ we have $S(t) \leq t$, and on $[3, x]$ we insert (10) and use Lemma 5.2 with $b = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$:

$$\int_1^x S(t) dt = \frac{x^2}{2\sqrt{\log x}} \left(\lambda_0 + \frac{\lambda_0/4 + \lambda_1}{\log x} \right) + O\left(\frac{x^2}{(\log x)^{5/2}} \right).$$

Subtracting this from $xS(x)$ gives

$$\sum_{n \leq x} \rho(n) = \frac{x^2}{2\sqrt{\log x}} \left(\lambda_0 + \frac{\lambda_1 - \lambda_0/4}{\log x} \right) + O\left(\frac{x^2}{(\log x)^{5/2}} \right).$$

Thus $C_2 = \lambda_0/2 = G(1)/(2\sqrt{\pi})$ and, by (6), $\kappa_2 = \lambda_1/\lambda_0 - \frac{1}{4} = \frac{1}{4} - \frac{\gamma}{4} - \frac{1}{2} G'(1)/G(1)$, which are the stated formulas. $\square$

The Euler product $G(1) = \prod_p (1 + \frac{1}{2p})(1 - \frac{1}{p})^{1/2} = 0.812105711163122\dots$ converges slowly, so we evaluated it, and the sum in κ_2 , through the prime zeta function $\mathcal{P}(s) = \sum_p p^{-s}$:

$$\log G(1) = \sum_{k \geq 2} \frac{(-1)^{k+1} 2^{-k} - \frac{1}{2}}{k} \mathcal{P}(k), \quad \sum_p \frac{\log p}{(2p+1)(p-1)} = -\frac{1}{3} \sum_{k \geq 0} \left(1 - \left(-\frac{1}{2}\right)^{k+1} \right) \mathcal{P}'(k+2).$$

Our value of $G(1)$ agrees with all 25 digits given in [4].

5.3 Numerical check

We computed $\mathcal{E}(x)$ exactly for $x \leq 2 \cdot 10^{10}$ with a bit sieve that removes the odd multiples of primes $p \equiv 1 \pmod{4}$, and $\sum_{n \leq x} \rho(n)$ exactly for $x \leq 10^{10}$ with a segmented sieve for $\varphi(n)$ and $\omega(n)$. Since $2\rho(n)$ is always an integer, the second sum was accumulated in exact 128-bit integer arithmetic.⁵ Tables 2 and 3 and Figures 3 and 4 compare the counts with the one-term and two-term predictions

$$\mathcal{E}_1(x) = \frac{Cx}{\sqrt{\log x}}, \quad \mathcal{E}_2(x) = \mathcal{E}_1(x) \left(1 + \frac{\kappa}{\log x} \right), \quad R_1(x) = \frac{C_2 x^2}{\sqrt{\log x}}, \quad R_2(x) = R_1(x) \left(1 + \frac{\kappa_2}{\log x} \right).$$

The one-term ratio $\mathcal{E}(x)/\mathcal{E}_1(x)$ is still 1.032 at $x = 2 \cdot 10^{10}$, which is the slow $1/\log x$ convergence one expects. The two-term prediction removes most of the gap. If the expansion is carried one step further, $\mathcal{E}(x)/\mathcal{E}_2(x) - 1 = \kappa_3(\log x)^{-2} + O((\log x)^{-3})$, and the same computation as in the proof of Theorem 1.3, with $Z''(1)$ and $\Gamma(-\frac{3}{2}) = \frac{4}{3}\sqrt{\pi}$, gives $\kappa_3 = 1.2639\dots$. The last column of Table 2 is consistent with this: over all our checkpoints $10^6 \leq x \leq 2 \cdot 10^{10}$, including those not shown in the table, it lies between 1.46 and 1.87, and apart from small fluctuations it falls from 1.74 at $x = 10^6$ to 1.46 at $x = 2 \cdot 10^{10}$, towards κ_3 . For the mean value κ_2 is small, and the one-term prediction is already within 0.13% at $x = 10^{10}$. The corresponding third coefficient is 0.0727..., and over all checkpoints $10^6 \leq x \leq 10^{10}$ the last column of Table 3 lies between 0.106 and 0.148 and falls from 0.139 at $x = 10^6$ to 0.106 at $x = 10^{10}$.

6 Conclusions

The ratio of surjective group homomorphisms to ring homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ is $2^{\varepsilon(n)} |U_n^2|$, with $\varepsilon(n)$ determined by the 2-part of n , and it is an integer exactly off the set $\mathcal{E}$ of $n \equiv 2 \pmod{4}$ whose odd part has only prime factors $\equiv 3 \pmod{4}$. For odd n the ring homomorphisms are in

⁵Both programs are short C files; the larger run took about nine minutes on one core. They are included in the archive listed in the data availability statement.

Table 2. The exceptional set: exact counts against Theorem 1.3.

x	$\mathcal{E}(x)$	$\mathcal{E}_1(x)$	$\mathcal{E}_2(x)$	$\mathcal{E}/\mathcal{E}_1$	$\mathcal{E}/\mathcal{E}_2$	$(\mathcal{E}/\mathcal{E}_2 - 1)(\log x)^2$
10^4	886	802	863	1.10535	1.02679	2.27
10^5	7729	7169	7608	1.07807	1.01588	2.10
10^6	69413	65447	68785	1.06061	1.00913	1.74
10^7	636229	605918	632411	1.05003	1.00604	1.57
10^8	5911404	5667842	5884687	1.04297	1.00454	1.54
10^9	55450285	53436931	55254205	1.03768	1.00355	1.52
10^{10}	523916546	506947235	522463396	1.03347	1.00278	1.47
$2 \cdot 10^{10}$	1031330156	998969959	1028651968	1.03239	1.00260	1.46

Table 3. The mean value of ρ : exact sums against Theorem 1.4.

x	$\sum_{n \leq x} \rho(n)$	$\sum \rho/R_1$	$\sum \rho/R_2$	$(\sum \rho/R_2 - 1)(\log x)^2$
10^4	$7.587531 \cdot 10^6$	1.005150	1.002447	0.208
10^5	$6.775787 \cdot 10^8$	1.003564	1.001404	0.186
10^6	$6.179019 \cdot 10^{10}$	1.002525	1.000727	0.139
10^7	$5.717876 \cdot 10^{12}$	1.002037	1.000496	0.129
10^8	$5.346803 \cdot 10^{14}$	1.001704	1.000356	0.121
10^9	$5.039785 \cdot 10^{16}$	1.001460	1.000261	0.112
10^{10}	$4.780294 \cdot 10^{18}$	1.001278	1.000200	0.106

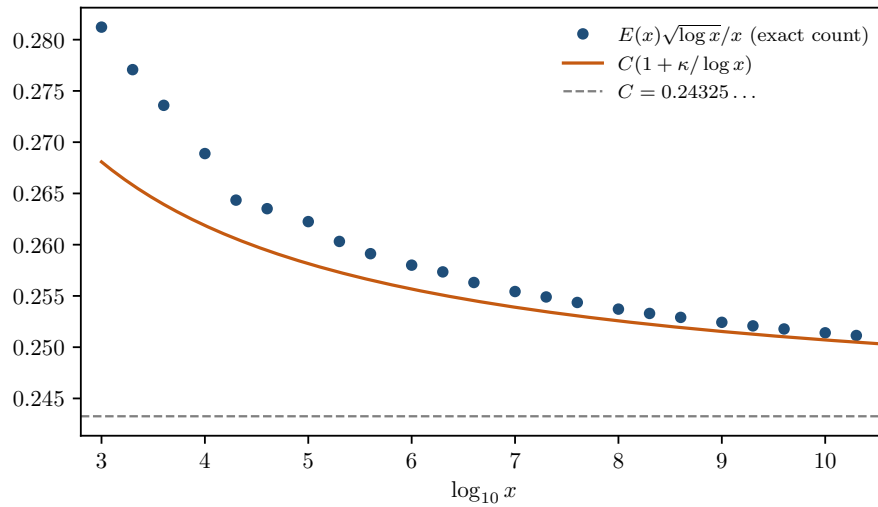


Figure 3. $\mathcal{E}(x)\sqrt{\log x}/x$ for $10^3 \leq x \leq 2 \cdot 10^{10}$, $x = c \cdot 10^k$ with $c \in \{1, 2, 4\}$, against the limit C and the two-term curve $C(1 + \kappa/\log x)$ of Theorem 1.3.

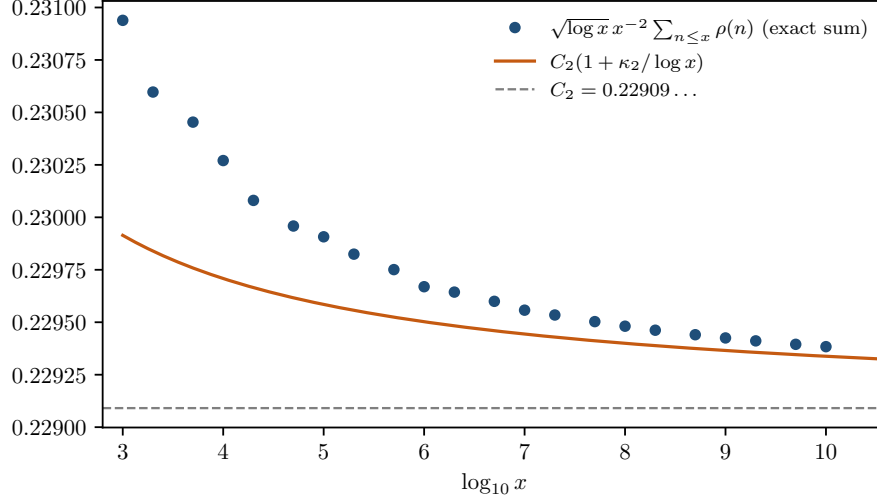


Figure 4. $x^{-2}\sqrt{\log x} \sum_{n \leq x} \rho(n)$ for $10^3 \leq x \leq 10^{10}$, $x = c \cdot 10^k$ with $c \in \{1, 2, 5\}$, against the limit C_2 and the two-term curve $C_2(1 + \kappa_2/\log x)$ of Theorem 1.4.

canonical bijection with the subfields of $\mathbb{Q}(\zeta_n)$ of degree at most 2, and $\rho(n)$ is the degree of $\mathbb{Q}(\zeta_n)$ over its maximal multiquadratic subfield. The set $\mathcal{E}$ has $Cx(\log x)^{-1/2}(1 + \kappa/\log x + O((\log x)^{-2}))$ elements up to x , and the partial sums of ρ are $C_2x^2(\log x)^{-1/2}(1 + \kappa_2/\log x + O((\log x)^{-2}))$, with all four constants explicit and confirmed by exact computation to $2 \cdot 10^{10}$ and 10^{10} respectively. The same method gives further terms of both expansions, at the cost of higher derivatives of $L(s, \chi_{-4})$ and of the Euler products at $s = 1$.

Author contributions

[To be completed by the authors.]

Use of AI tools declaration

[To be completed by the authors.]

Acknowledgments

The authors thank Prof. Jaikumar Radhakrishnan (ICTS-TIFR, Bengaluru) for helpful comments.

Funding

The authors did not receive any financial support for the research, authorship, or publication of this article.

Conflict of interest

The authors declare no conflicts of interest.

Data availability

The C and Python programs and the exact counts behind Tables 2 and 3 are available at <https://github.com/creelie/quadratic-residue>.

References

- [1] J. A. Gallian, J. Van Buskirk, The number of homomorphisms from $\mathbb{Z}_m$ into $\mathbb{Z}_n$, *Amer. Math. Mon.*, **91** (1984), 196–197. <https://doi.org/10.1080/00029890.1984.11971375>
- [2] P. Moree, Chebyshev’s bias for composite numbers with restricted prime divisors, *Math. Comp.*, **73** (2004), 425–449. <https://doi.org/10.1090/S0025-5718-03-01536-0>
- [3] G. A. Jones, A. K. Zvonkin, A number-theoretic problem concerning pseudo-real Riemann surfaces, *Integers*, **25** (2025), A75. <https://doi.org/10.5281/zenodo.16881820>
- [4] S. Finch, P. Sebah, Squares and cubes modulo n , *arXiv:math/0604465v3*, 2016. <https://doi.org/10.48550/arXiv.math/0604465>
- [5] G. Tenenbaum, *Introduction to analytic and probabilistic number theory*, 3 Eds., Providence: American Mathematical Society, 2015. <https://doi.org/10.1090/gsm/163>
- [6] K. Ireland, M. Rosen, *A classical introduction to modern number theory*, 2 Eds., New York: Springer, 1990. <https://doi.org/10.1007/978-1-4757-2103-4>
- [7] L. C. Washington, *Introduction to cyclotomic fields*, 2 Eds., New York: Springer, 1997. <https://doi.org/10.1007/978-1-4612-1934-7>
- [8] H. L. Montgomery, R. C. Vaughan, *Multiplicative number theory I: Classical theory*, Cambridge: Cambridge University Press, 2006. <https://doi.org/10.1017/CB09780511618314>